

CSC 365: Internet Programming
Homework 4: Cheek Hall Classroom Reservation System

1. Goal: Practice using React.

2. Requirements

- (a) Download the start code from BrightSpace.
- (b) Create a React component named *Cheekhall_List* that stores the availability of all classrooms (true = available, false = reserved).
- (c) Create a React component named *Department_List* that tracks the classrooms reserved by a specific department (e.g., Computer Science or Mathematics).
- (d) In *Department_List.jsx*, implement a function to reserve a classroom in Cheekhall. When a room is reserved, update the shared Cheekhall state so that the room's status is set to false.
- (e) Provide reservation buttons for each department for every classroom.
- (f) If a department attempts to reserve a classroom that is already reserved, display an alert.

3. Demo

- (a) Initially, your website should look like the example as shown in Figure 1.
- (b) When a classroom is reserved, the website should look like the example as shown in Figure 2.
- (c) If a department attempts to reserve an already-reserved classroom, the website should display an alert as shown in Figure 3.

Welcome to the Classroom reserve list!

Available Classrooms

- 205: Available
- 206: Available
- 207: Available
- 208: Available
- 209: Available

Classrooms reserved —

- 205: —
- 206: —
- 207: —
- 208: —
- 209: —

Reserve 205	Reserve 206	Reserve 207	Reserve 208	Reserve 209
-------------	-------------	-------------	-------------	-------------

Classrooms reserved —

- 205: —
- 206: —
- 207: —
- 208: —
- 209: —

Reserve 205	Reserve 206	Reserve 207	Reserve 208	Reserve 209
-------------	-------------	-------------	-------------	-------------

Figure 1: The illustration of the beginning of the web page.

Welcome to the Classroom reserve list!

Available Classrooms

- 205: Available
- 206: Reserved
- 207: Reserved
- 208: Available
- 209: Reserved

Classrooms reserved —

- 205: —
- 206: Reserved
- 207: Reserved
- 208: —
- 209: —

Reserve 205	Reserve 206	Reserve 207	Reserve 208	Reserve 209
-------------	-------------	-------------	-------------	-------------

Classrooms reserved —

- 205: —
- 206: —
- 207: —
- 208: —
- 209: Reserved

Reserve 205	Reserve 206	Reserve 207	Reserve 208	Reserve 209
-------------	-------------	-------------	-------------	-------------

Figure 2: The illustration of the webpage after reservation

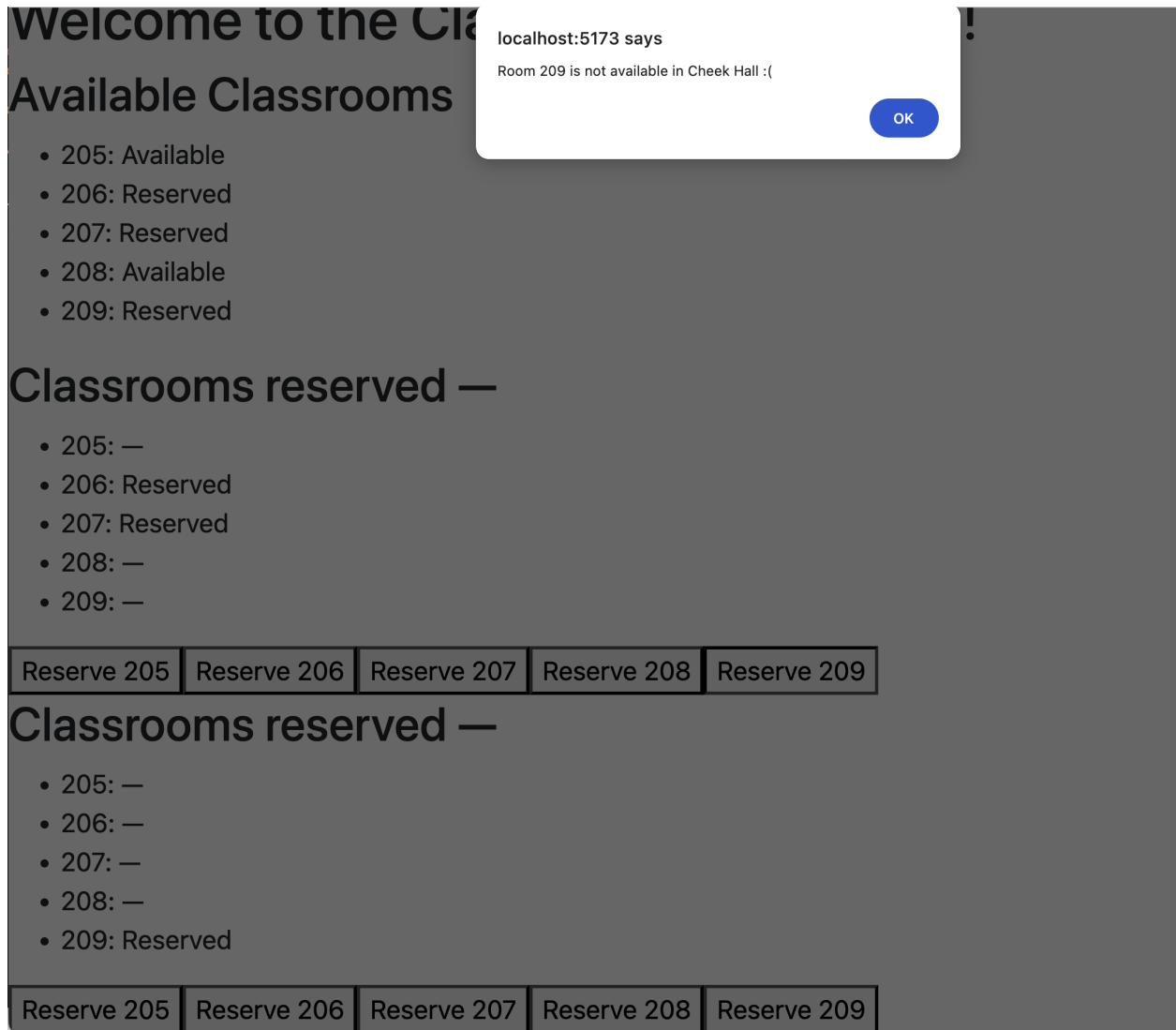


Figure 3: The illustration of the alert